Introduction to Feedback Control – Open-loop and Close-loop Control

(反馈控制介绍 – 开环和闭环控制)

Object of Control System Design

- **Factors Influencing Output of A Dynamic System (影响动态系统输出的因素)**

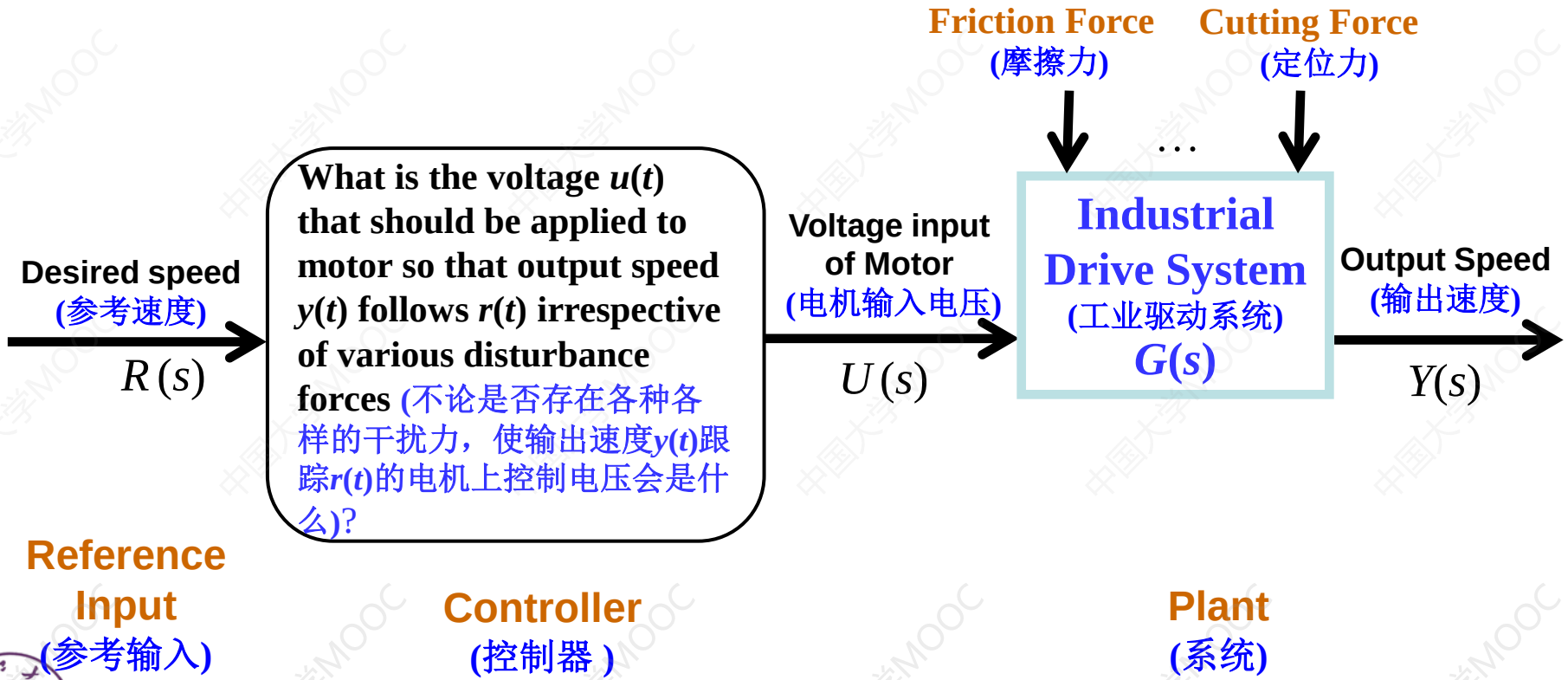
Various factor that cannot be predicted in the design of a product (e.g., slope of road and wind forces) (设计中不能被预测的因素(如路面斜坡和风力))

→

The system may not possess the quality that one would like to have (e.g., chattering problem of fast car braking action) (系统可能不具有期望的品质(如快速运动的汽车制动时的抖动问题))

- **Objective of A Control System Design (控制系统设计的目标)**
  - Figure out strategy to adjust certain inputs to the system (e.g., the gas pedal position) so that the output of the system behaves in the way one would like to have (e.g., follow the desired reference output) in spite of various factors that cannot be predicted in the design of a product (尽管设计中有各种各样不能预测的因素，我们需要设计出控制策略来调节系统的输入(如油门踏板位置)，从而使系统的输出行为与期望一致(如跟踪期望的参考输出))

Object of Control System Design



Open-Loop vs Closed-Loop

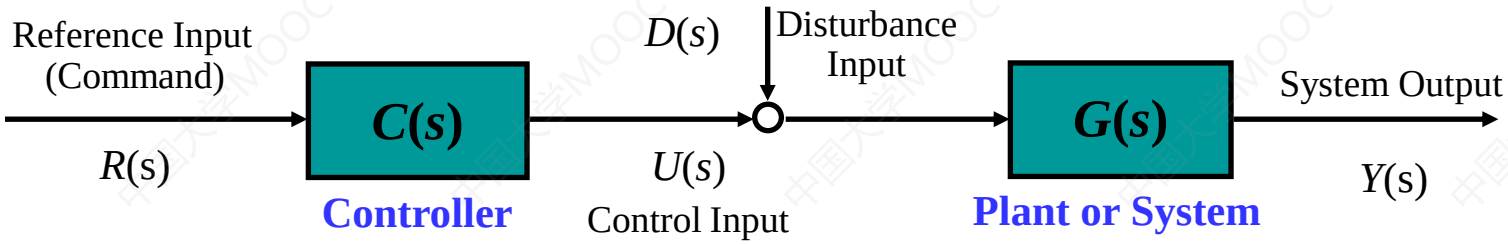
- **Open-Loop Control (开环控制)**

  - The control input $u(t)$ (or $U(s)$) is synthesized based on the priori knowledge of the physical system (plant) and the reference input r(t) (or $R(s)$) **(基于对物理系统的已有认识和参考输入$r(t)$来设计控制量$u(t)$)**

  - The control system does not measure the output, and there is no comparison of the actual output to the reference input when generating control input **(控制系统不需要检测输出，当生成控制输入时实际输出和参考输入没有对比)**

Open-Loop vs Closed-Loop

- **Open-Loop Control (开环控制)**



Q:Ideally, if we want Y(s) to follow R(s) (i.e. want Y(s) = R(s)), how would you design the controller C(s) for the above open-loop control system? **(理想情况，如果我们想要*Y*(*s*)跟踪*R*(*s*),你会如何设计上述开环控制系统的控制器*C*(*s*)？)**

Want **(希望):**

$$Y(s) = G(s)[u(s) + D(s)]$$
$$= G(s)C(s)R(s) + G(s)D(s) = R(s)$$

$$\Rightarrow \quad C(s) = G^{-1}(s)$$

System inversion! (系统求逆)

Q:Can we attenuate the effect of disturbance D(s) on system output Y(s)? **(我们能抑制干扰*D*(*s*)对系统输出*Y*(*s*)的影响吗？)**

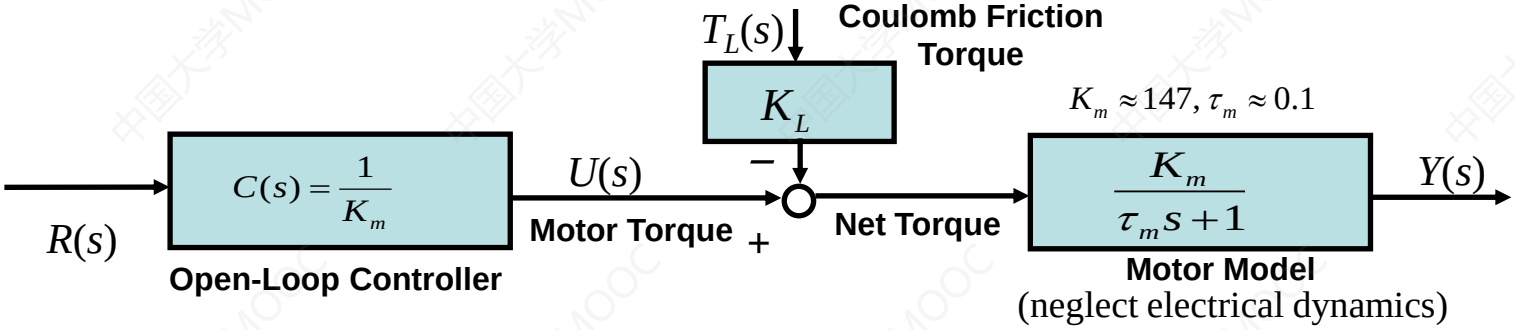
No!

Q:Can we attenuate the effect of variations of plant transfer function G(s) on system output Y(s)? **(我们能抑制系统传递函数*G*(*s*)变化对系统输出*Y*(*s*)的影响吗？)**

No!

Open-Loop Control Example

- **Speed Control of a DC Motor without Sensors (无传感器的直流电机速度控制)**
  - ➢**Block Diagram of Open-Loop Control System (开环控制系统的方框图)**



➢**Ideal Open-Loop Controller (理想开环控制器)**

$$C_{\text{ideal}}(s) = \frac{1}{G(s)} = \frac{\tau_m s + 1}{K_m}$$

$$\Rightarrow U(s) = C_{\text{ideal}}(s)R(s) = \frac{\tau_m}{K_m} sR(s) + \frac{1}{K_m} R(s)$$

In time-domain: $u(t) = \frac{\tau_m}{K_m}\dot{r}(t) + \frac{1}{K_m} r(t)$

➢**An Implementable Open-Loop Controller (可实施的开环控制器)**

$$C(s) = \frac{1}{K_m} \quad \Rightarrow \quad U(s) = \frac{1}{K_m} R(s) \Leftrightarrow u(t) = \frac{1}{K_m} r(t)$$

➢**Output Speed with Open-Loop Controller (开环控制器的输出速度)**

$$Y(s) = \frac{\cancel{K_m}}{\cancel{K_m}(\tau_m s + 1)} R(s) + \left[-\frac{K_L K_m}{\tau_m s + 1} \right] T_L(s)$$

Open-Loop Control Example

- **Speed Control of a DC Motor without Sensors** (无传感器的直流电机速度控制)
  - ➢**Steady-state output speed for constant desired speed reference inputs of *r* (no friction)** (恒定期望参考速度输入下的稳态输出速度(无摩擦))

$$T_L(s)=0 \Rightarrow Y(s)=\frac{1}{(\tau_m s+1)}R(s)$$

For constant desired speed (恒定期望速度) $R(s)=r/s$

$$y_{ss}=\lim_{s\to 0} sY(s)=\lim_{s\to 0}\frac{1}{\tau_m s+1}sR(s)=r$$

Time constant (no controller) (有控制器的时间常数) $\boxed{\tau_m}$

Time constant (with controller) (无控制器的时间常数) $\boxed{\tau_m}$

No Change!

  - ➢**Steady-state output speed for constant desired speed reference inputs of *r* (with a constant friction τ_L)** (恒定期望参考速度输入下的稳态输出速度(恒定摩擦))

$$y_{ss}=\lim_{s\to 0} sY(s)=\lim_{s\to 0}\left[\frac{1}{(\tau_m s+1)}sR(s)-\frac{K_m K_L}{\tau_m s+1}sT_L(s)\right]$$

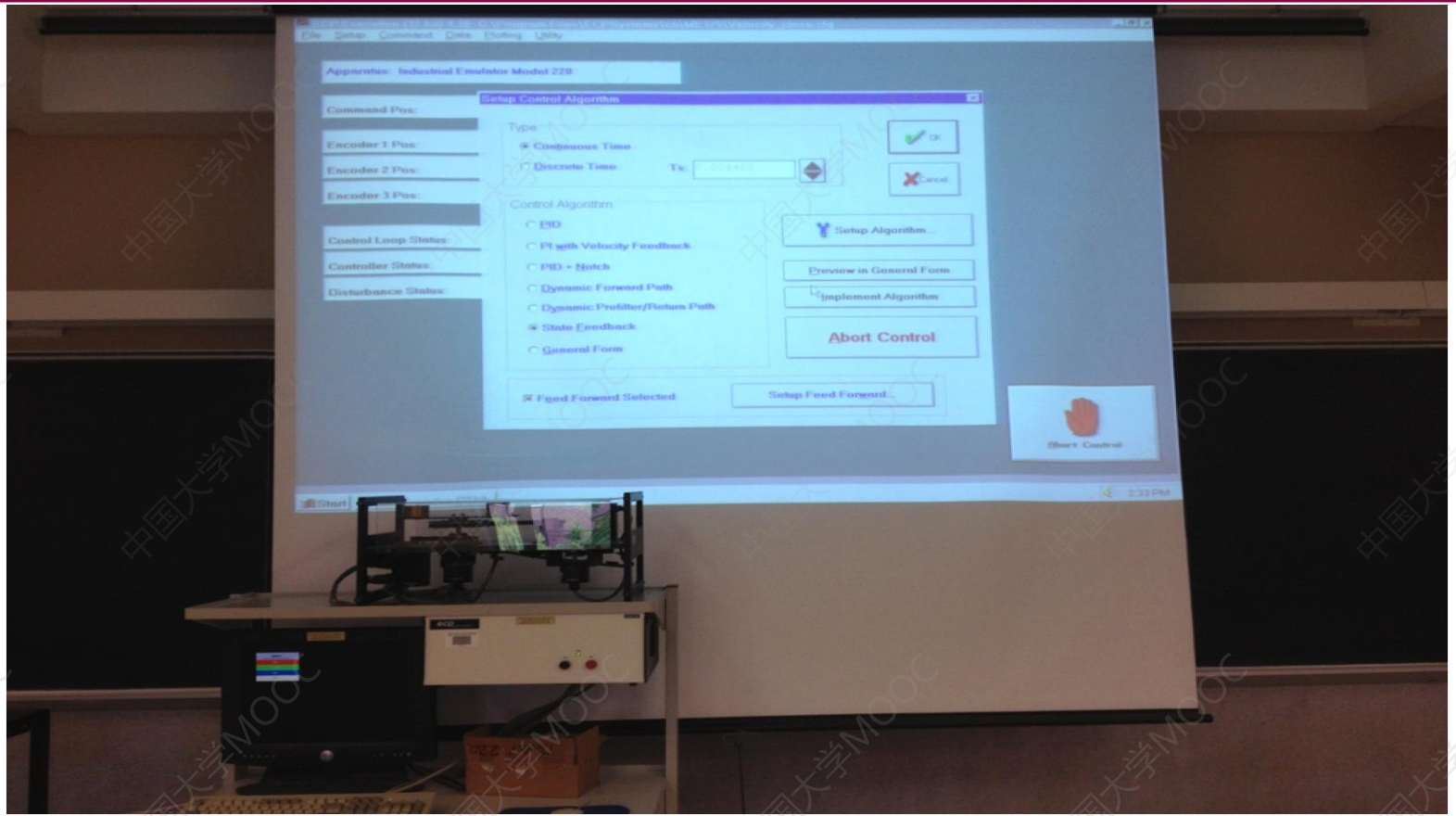
$$=r-K_m K_L \tau_L$$

There is no attenuation on the effect of friction ! (摩擦影响没有得到抑制!)

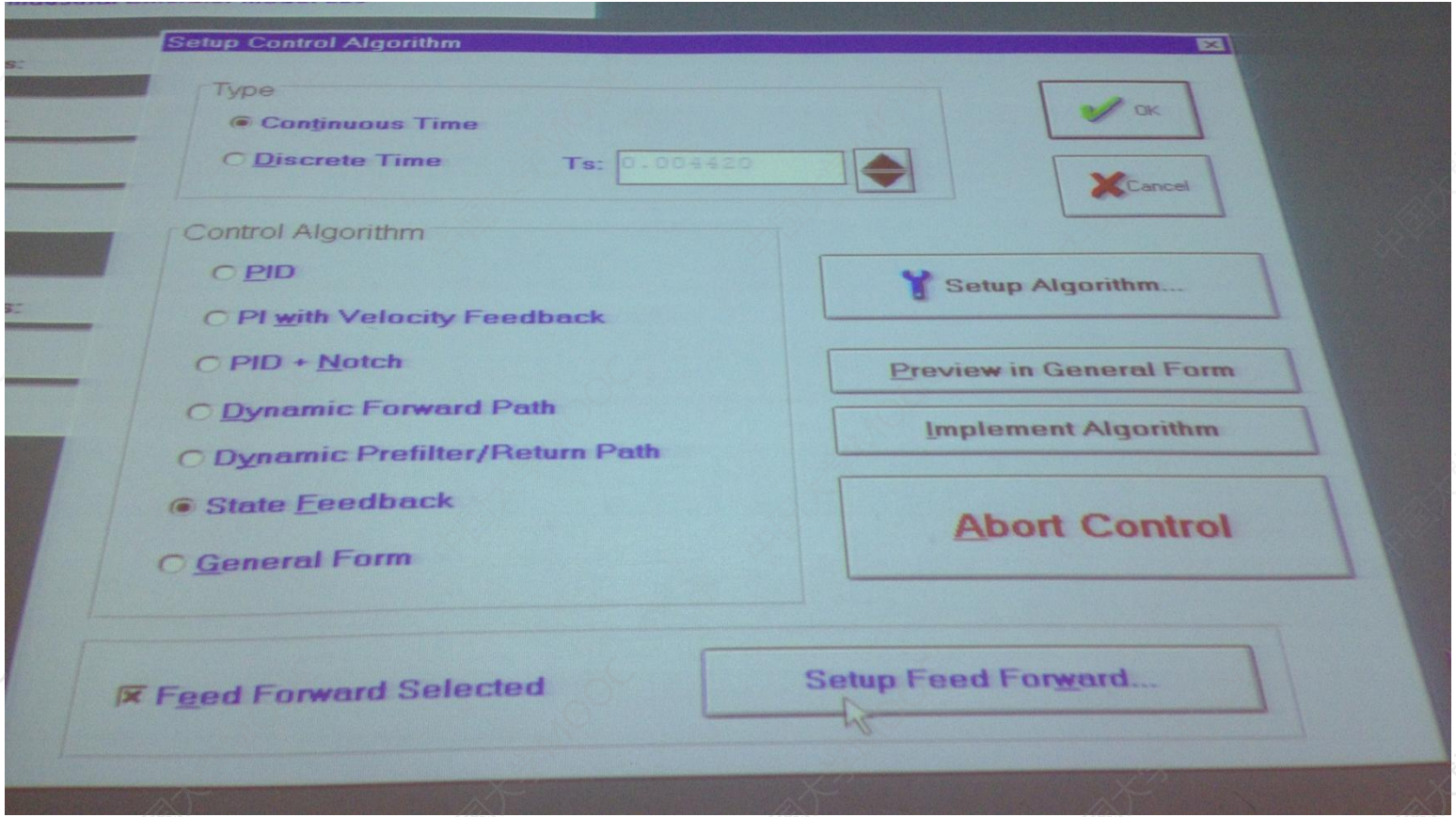
Q: In reality, the friction on a motor may change quite significantly. Will the customer be happy with such an open-loop controller ? (实际电机摩擦可能有显著改变，用户会满意这种开环控制器吗?)

No!

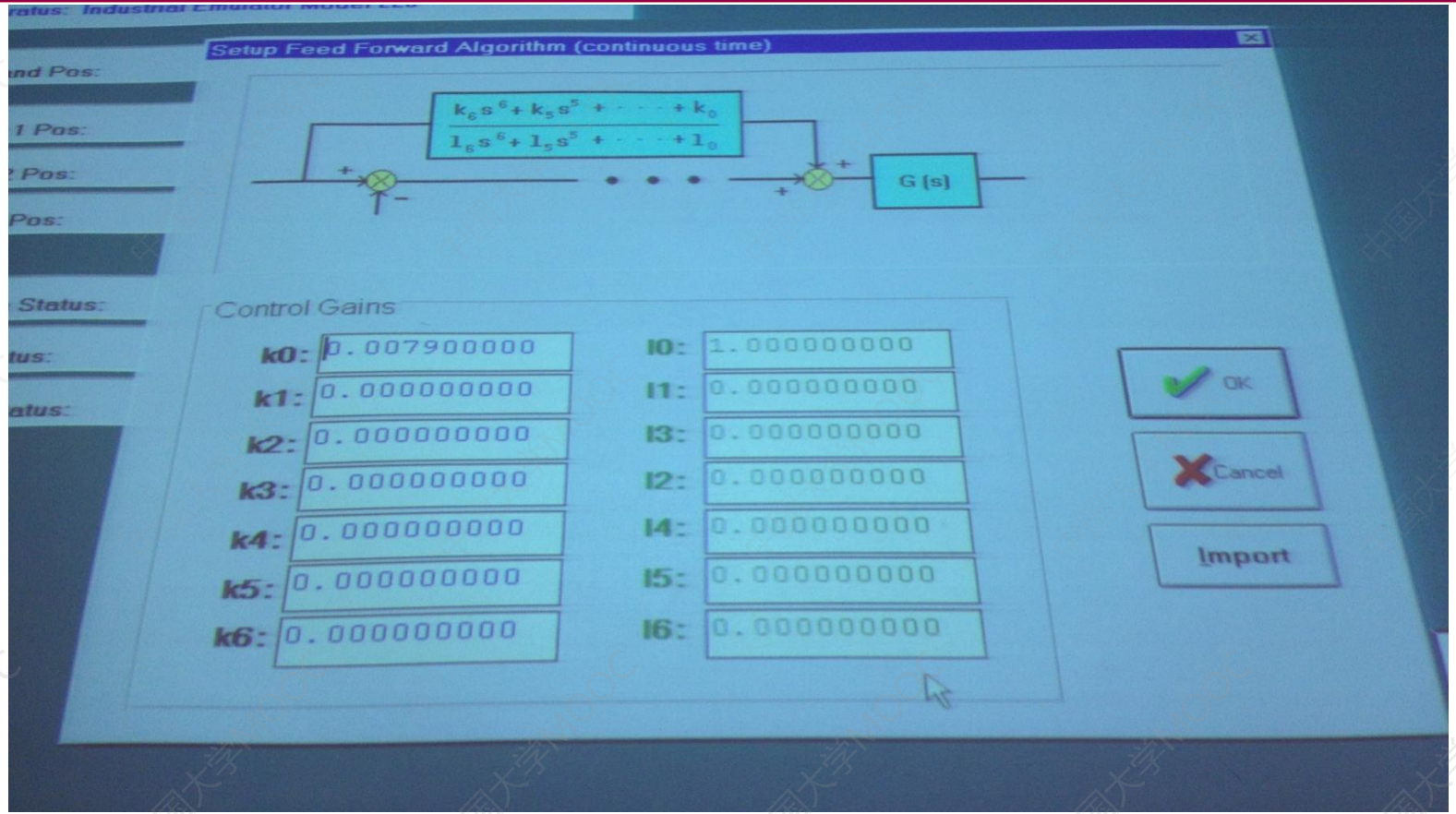
Experiments on Speed Control of a DC Motor



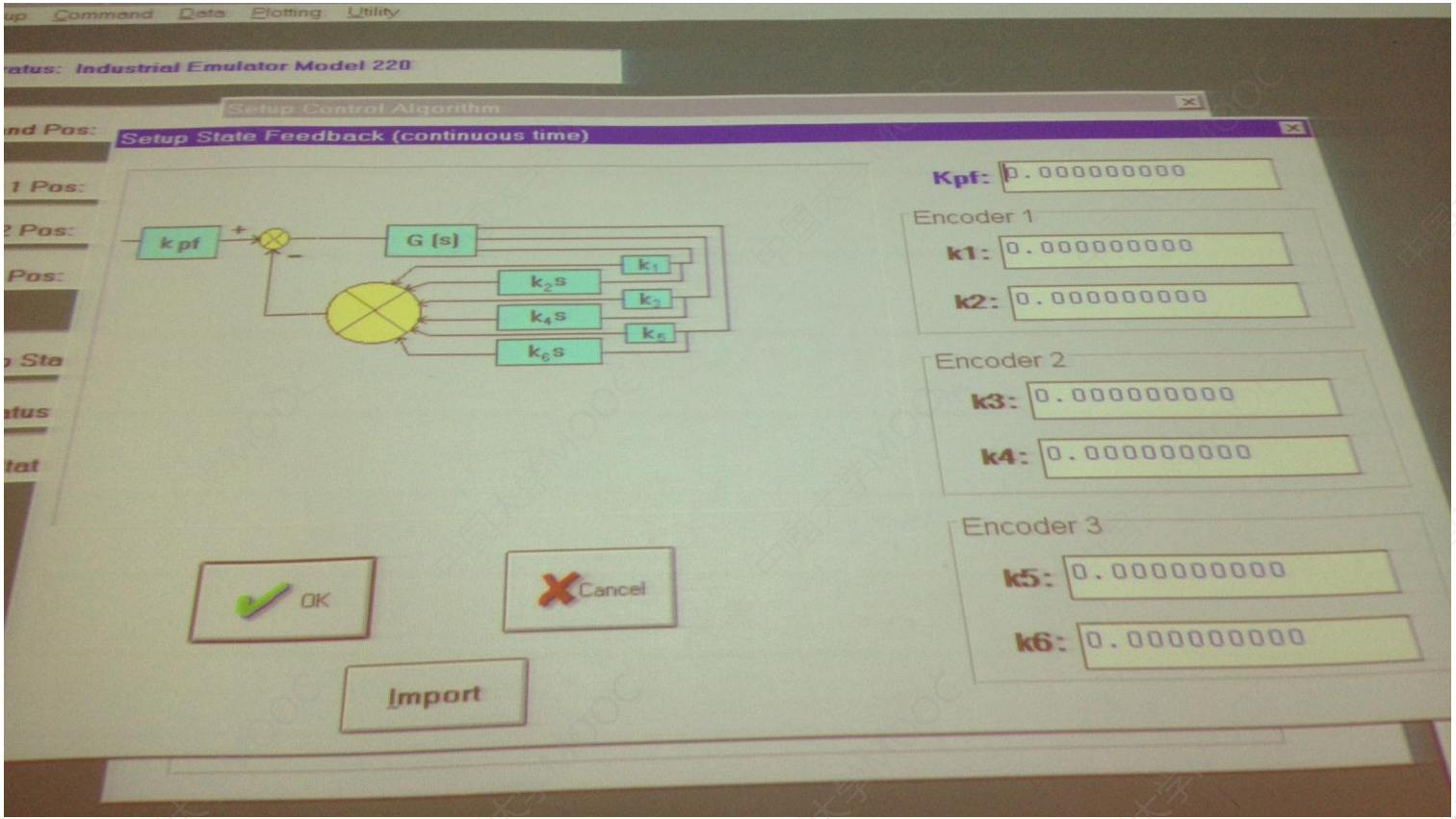
Experiments on Speed Control of a DC Motor



Experiments on Speed Control of a DC Motor

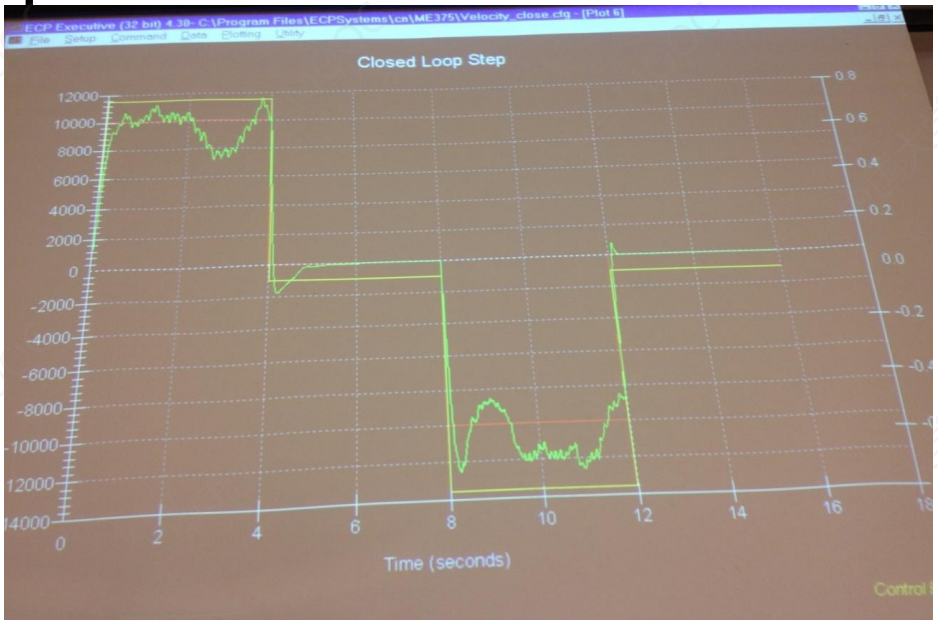


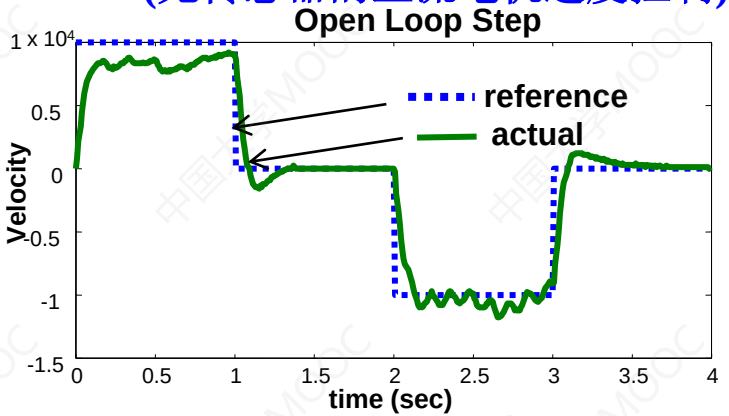
Experiments on Speed Control of a DC Motor

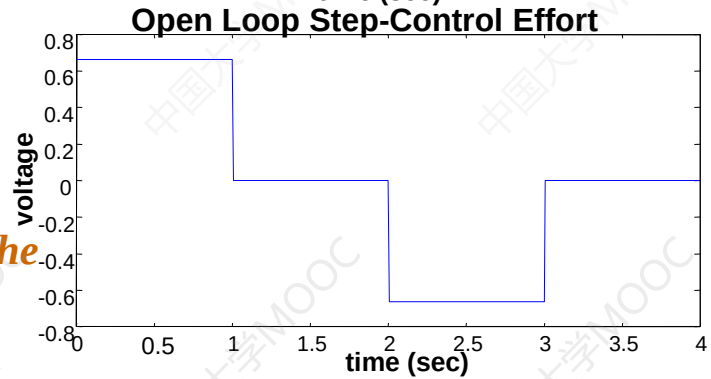


Open Loop Experimental Results

- **Speed Control of a DC Motor without Sensors** (无传感器的直流电机速度控制)



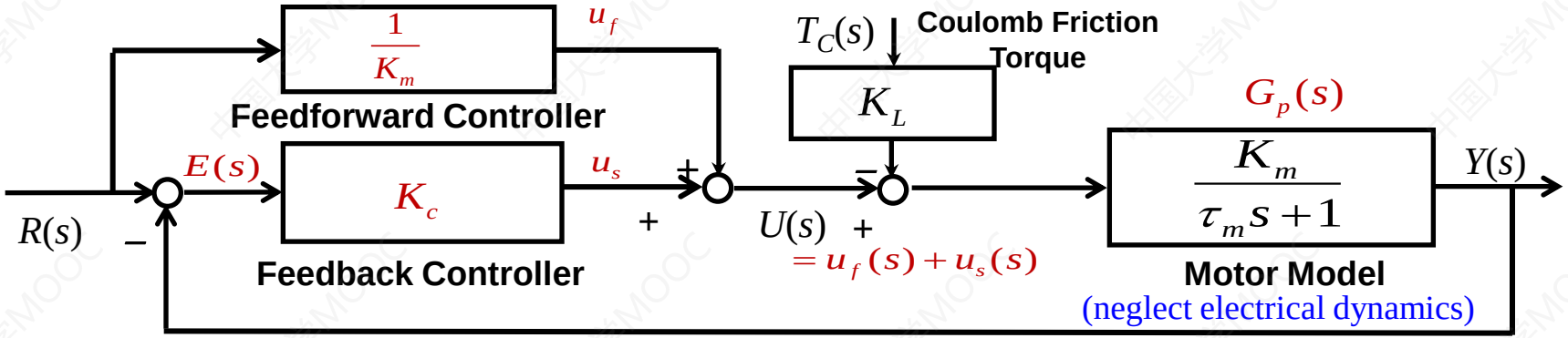




Q: Can you explain the experimental results obtained (e.g., the quality of the tracking performance and how it is different from the theoretical prediction)? (你能解释所获得的实验结果吗(跟踪性能的品质和其与理论预测的不同之处)?)

Closed-Loop Control Example

- **Speed Control of a DC Motor with Speed Sensors**
  (有传感器的直流电机速度控制)



Block Diagram of A Simple Closed-Loop Control System

- **A Simple Closed-Loop Controller** (一个简单闭环控制器)
  - **Feedforward Controller** (前馈控制器)

$$u_f(s) = \frac{1}{K_m} R(s) \quad \Leftrightarrow \quad u_f(t) = \frac{1}{K_m} r(t)$$

  - **Feedback Controller** (反馈控制器)

$$u_s(s) = K_c E(s) = K_c[R(s) - Y(s)] \quad \Leftrightarrow \quad u_s(t) = K_c[r(t) - y(t)]$$

Closed-Loop Control Example

- **Speed Control of a DC Motor with Speed Sensors**

  (有传感器的直流电机的速度控制)

  ➢**Output speed with closed-loop controller** (闭环控制器的输出速度)

$$Y(s)=\underbrace{\frac{1+K_mK_c}{\tau_m s+1+K_mK_c}}_{G_{YR}(s)}R(s)-\frac{K_LK_m}{\tau_m s+1+K_mK_c}T_L(s)$$

$$Y(s)=\underbrace{\frac{1}{\frac{\tau_m}{1+K_mK_c}s+1}}_{G_{YR}(s)}R(s)+\underbrace{\left[-\frac{\frac{K_LK_m}{1+K_mK_c}}{\frac{\tau_m}{1+K_mK_c}s+1}\right]}_{G_{YT}(s)}T_L(s)$$

Closed-loop time constant (闭环时间常数):

$$\boxed{\tau=\frac{\tau_m}{1+K_mK_c}}$$

➢**Steady-state (SS) output speed for constant desired speed reference inputs of *r* (no friction)** (无摩擦时的恒定期望速度参考输入 *r*下的稳态输出速度)

$$T_L(s)=0 \quad\Longrightarrow\quad y_{ss}=G_{YR}(0)r=r$$

No SS speed error ! (无静态速度误差)

➢**Steady-state output speed for constant desired speed reference inputs *r* (with a constant friction τ_L)** (有恒定摩擦力矩τ_L时的恒定期望速度参考输入 *r*下的稳态输出速度)

$$y_{ss}=G_{YR}(0)r+G_{YT}(0)\tau_L=r-\frac{K_LK_m}{1+K_mK_c}\tau_L$$

Closed-Loop Control Example

- **Speed Control of a DC Motor with Speed Sensors**

  (有传感器的直流电机的速度控制)

➢**Steady-state output speed error for constant desired speed reference inputs of *r* (with a constant friction τ_L) (有恒定摩擦力矩τ_L时的恒定期望速度参考输入 *r*下的稳态输出速度误差)**

$$e_{ss} = r - y_{ss} = \frac{K_L K_m}{1 + K_m K_c}\tau_L$$

decrease with increasing K_c!
(随K_c增大而降低)

***Q: When can we consistently have the desired steady-state speed regardless certain amount of Coulomb friction that may exist ?* (什么时候能得的一致的稳态速度而不管可能存在的摩擦力影响 ？)**

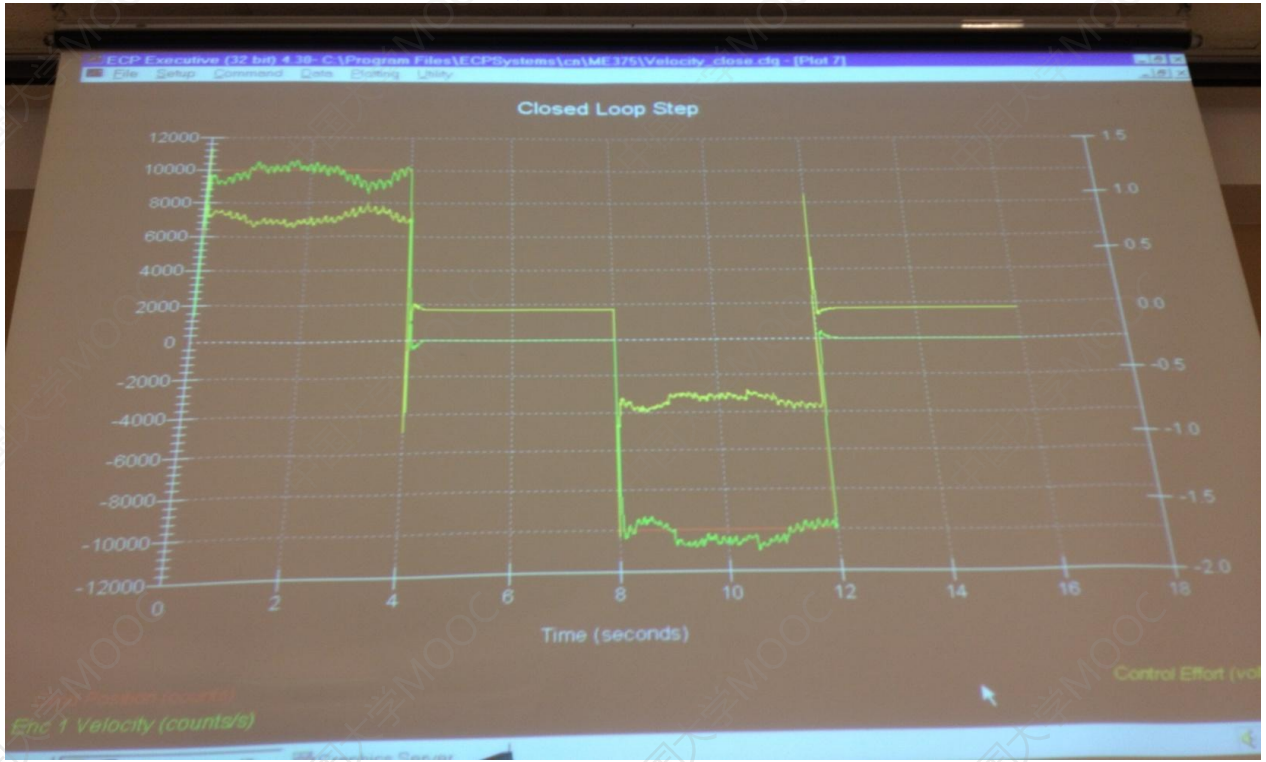
Using large enough feedback gain K_c so that the effect of disturbances like the Coulomb friction is attenuated sufficiently **(采用足够大的反馈增益K_c以便如库仑摩擦力之类的干扰影响能被有效抑制)**

***Q: How is response speed of the closed-loop system compared with the response speed of original open-loop system ?* (与原始开环系统相比，闭环系统的响应速度如何？)**

Faster as the CL time constant $\tau = \frac{\tau_m}{1 + K_m K_c}$ is smaller than open-loop time constant τ_m**(响应更快，由于闭环时间常数比开环时间常数小一些）**

Closed-Loop Experimental Results

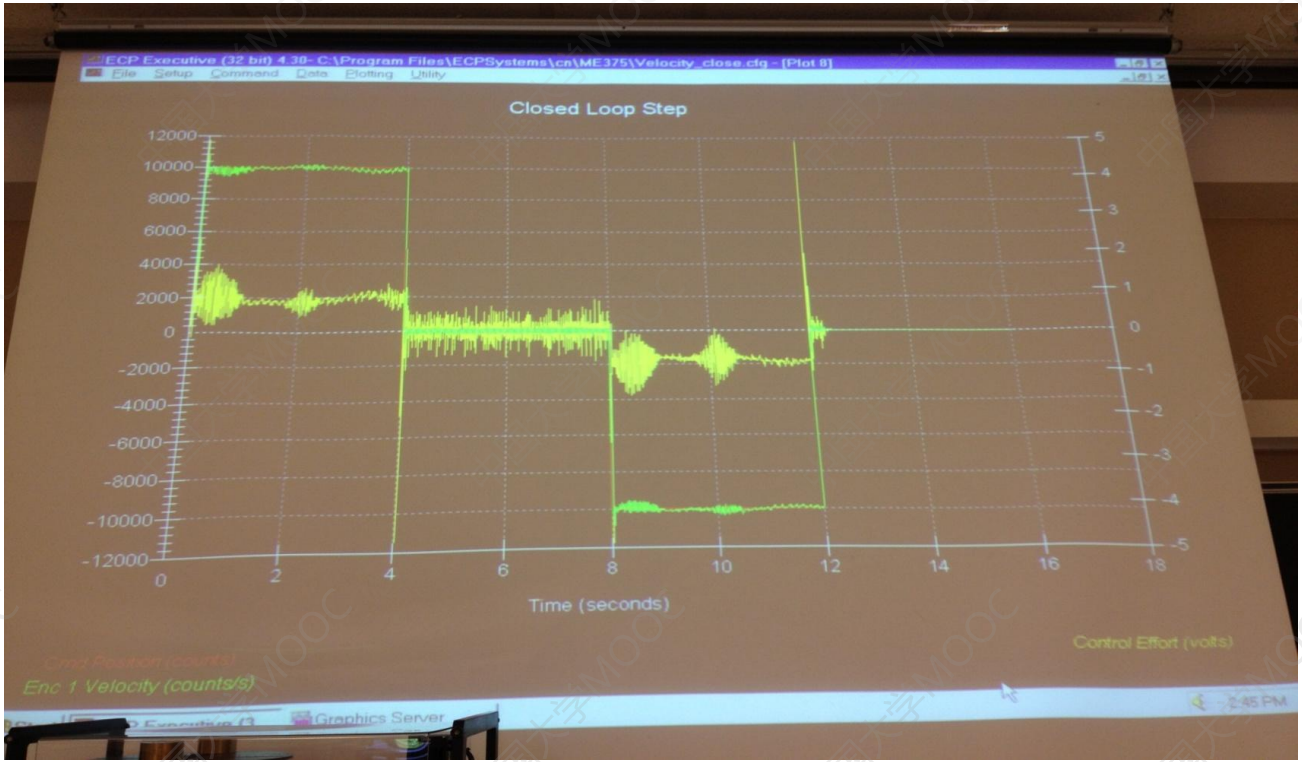
➢ **Speed Control of a DC Motor with Sensors and Feedforward**
(有传感器和前馈控制的直流电机速度控制)



K_c=0.01

Closed-Loop Experimental Results

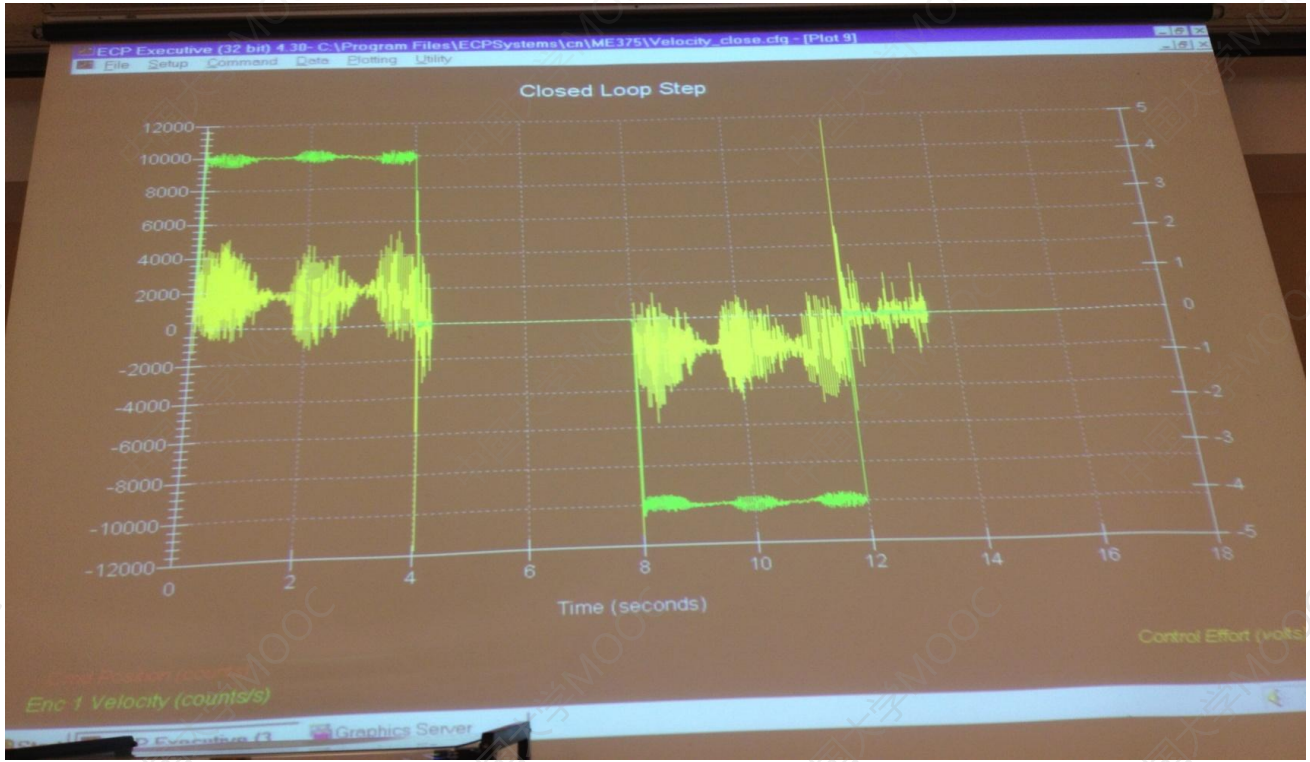
➢ **Speed Control of a DC Motor with Sensors and Feedforward**
(有传感器和前馈控制的直流电机的速度控制)



K_c=0.07

Closed-Loop Experimental Results

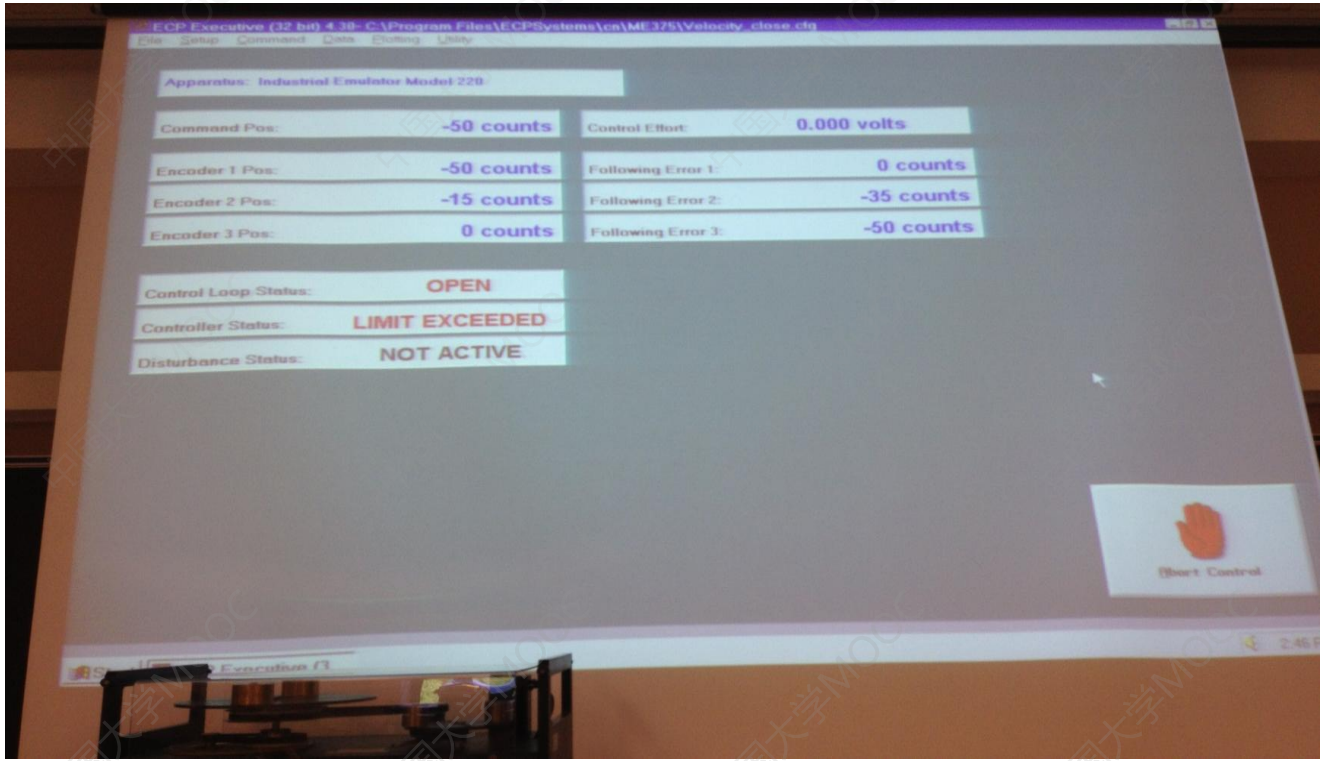
➢ **Speed Control of a DC Motor with Sensors and Feedforward**
(有传感器和前馈控制的直流电机的速度控制)



K_c=0.11

Closed-Loop Experimental Results

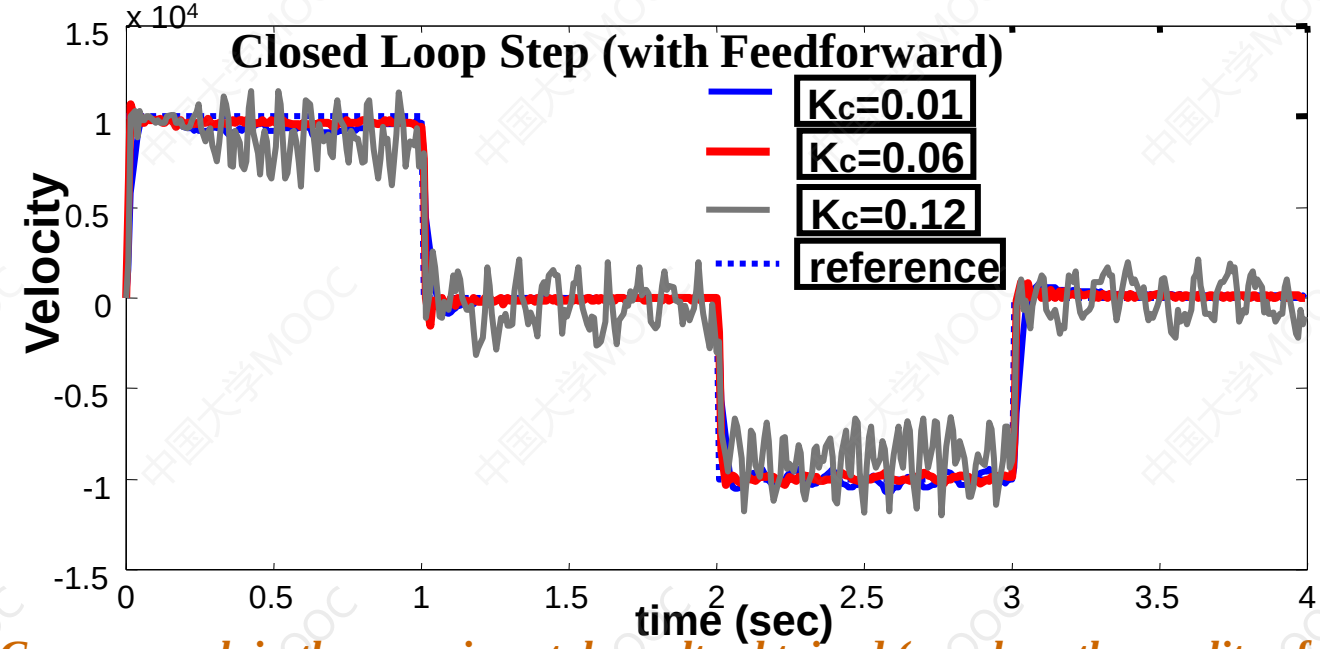
➢ **Speed Control of a DC Motor with Sensors and Feedforward**
(有传感器和前馈控制的直流电机的速度控制)



K_c=0.13

Closed-Loop Experimental Results

➢ **Speed Control of a DC Motor with Sensors and Feedforward**
(有传感器的直流电机的速度控制)

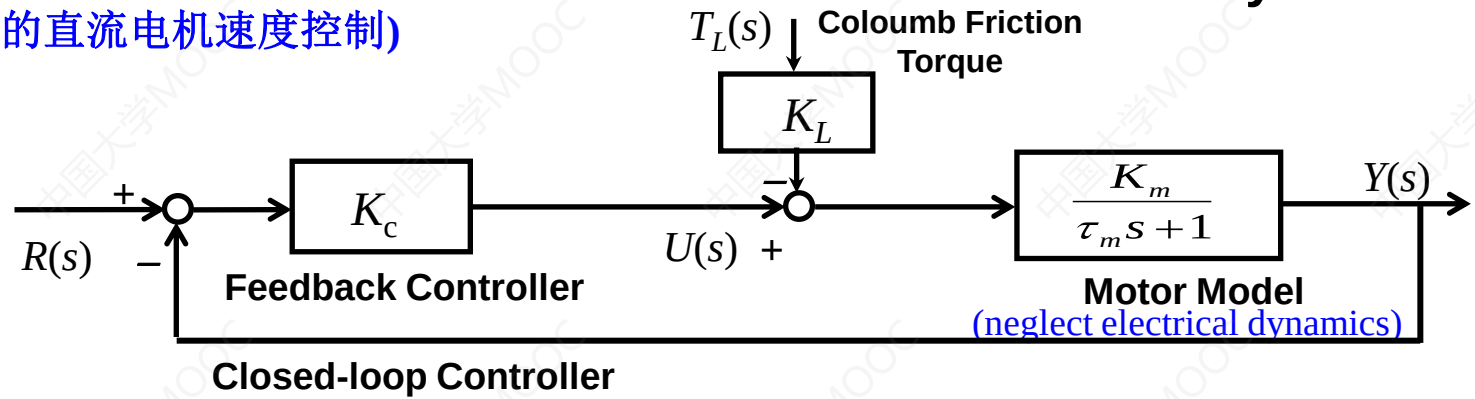


Q: Can you explain the experimental results obtained (e.g., how the quality of the tracking performance and response speed change with feedback gain as predicted by theory)? **(你能解释所获得的实验结果吗(跟踪性能的品质和由理论预测的响应速度变化随反馈增益如何变化？))**

Closed-Loop Control Example

- **Speed Control of a DC Motor with Feedback Action Only**

(仅有反馈的直流电机速度控制)



Block Diagram of A Simple Closed-Loop Control System (No feedforward)

$$Y(s)=\underbrace{\frac{K_m}{\tau_m s+1}}_{G_p(s)}\left[U(s)-K_L T_L(s)\right]=G_p(s)\left[K_c R(s)-K_c Y(s)-K_L T_L(s)\right]$$

$$\Rightarrow \quad Y(s)=\frac{G_p(s)K_c}{1+K_c G_p(s)}R(s)+\left[\frac{-K_L G_p(s)}{1+K_c G_p(s)}T_L(s)\right]=\frac{\frac{K_m}{\tau_m s+1}K_c}{1+K_c\frac{K_m}{\tau_m s+1}}R(s)-\frac{K_L\frac{K_m}{\tau_m s+1}}{1+K_c\frac{K_m}{\tau_m s+1}}T_L(s)$$

$$E(s)=R(s)-Y(s)=\frac{\frac{1}{1+K_m K_c}}{\frac{\tau_m}{1+K_m K_c}s+1}R(s)+\frac{\frac{K_m K_L}{1+K_m K_c}}{\frac{\tau_m}{1+K_m K_c}s+1}T_L(s)$$

$$y_{ss}=G_{YR}(0)r+G_{YT}(0)\tau_L=\frac{K_m K_c}{1+K_m K_c}r-\frac{K_L K_m}{1+K_m K_c}\tau_L$$

$$e_{ss}=r-y_{ss}=\frac{1}{1+K_m K_c}r+\frac{K_L K_m}{1+K_m K_c}\tau_L$$

Feedforward vs Feedback Control

- **Speed Control of a DC Motor with Sensors(有传感器的直流电机速度控制)**

➢**With Feedback and Feedforward (反馈和前馈控制):**

$$Y(s)=\frac{1}{\frac{\tau_m}{1+K_mK_c}s+1}R(s)-\frac{\frac{K_LK_m}{1+K_mK_c}}{\frac{\tau_m}{1+K_mK_c}s+1}T_L(s)$$

$$e_{ss}=r-y_{ss}=\frac{K_LK_m}{1+K_mK_c}\tau_L$$

➢**With Feedback Action Only (仅反馈控制):**

$$Y(s)=\frac{\frac{K_mK_c}{1+K_mK_c}}{\frac{\tau_m}{1+K_cK_m}s+1}R(s)-\frac{\frac{K_mK_L}{1+K_cK_m}}{\frac{\tau_m}{1+K_cK_m}s+1}T_L(s)$$

$$e_{ss}=r-y_{ss}=\frac{1}{1+K_mK_c}r+\frac{K_LK_m}{1+K_mK_c}\tau_L$$

Q: How is the performance of the closed-loop control with feedback control action only compared to that of the closed-loop control with both feedforward and feedback actions ? **(相比于同时具有前馈和反馈控制的闭环系统，仅有反馈控制的闭环系统的性能如何?)**

➢**Attenuation ability of the system to disturbances such as the Coulomb friction (系统对库仑摩擦等干扰的抑制能力):** **Same (相同)**

➢**Response speed (响应速度):**

$$\tau_m'=\frac{\tau_m}{1+K_mK_c}$$

Same (相同)

➢**Steady-state output speed error for unit constant desired speed without considering friction effect (不考虑摩擦影响时的单位定常期望速度的稳态输出速度误差):**

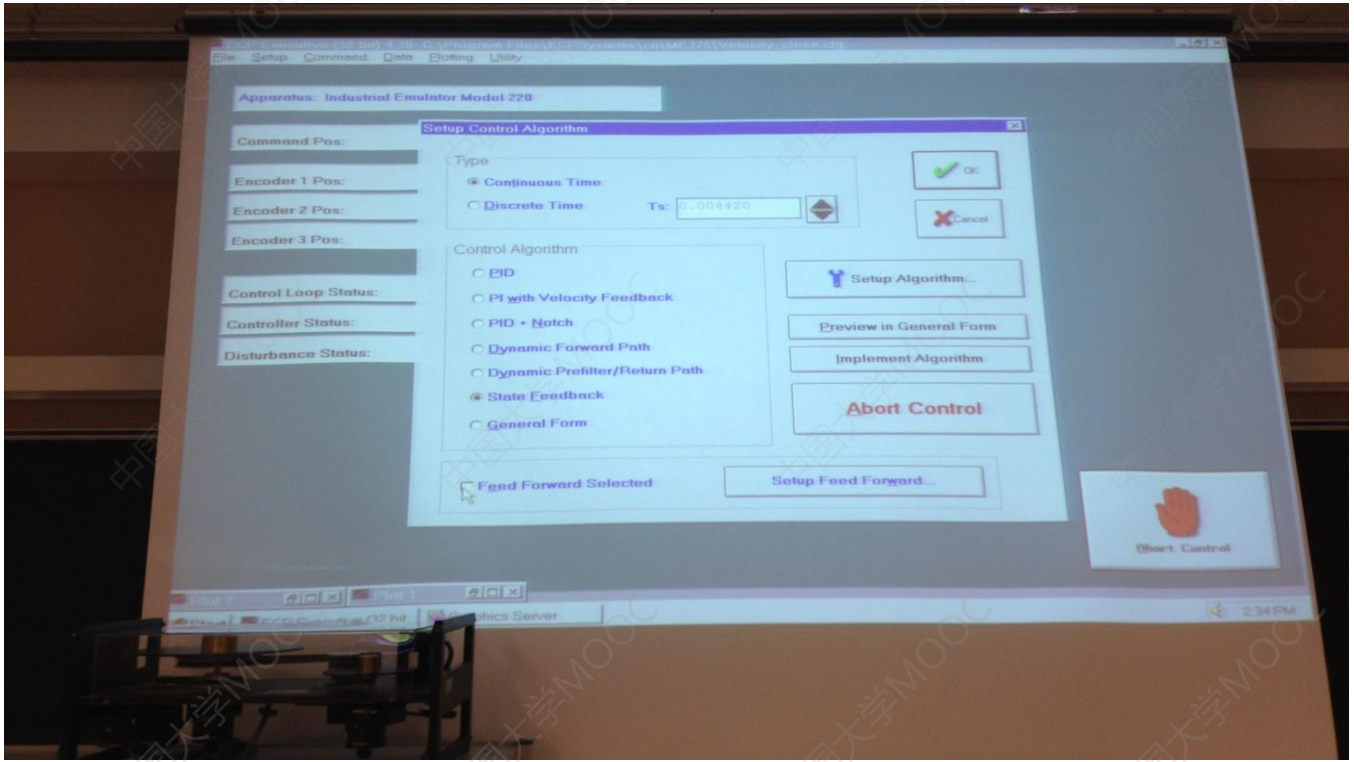
A little bit larger for Feedback Only (仅反馈控制时有点大)

when K_c is very large (当K_c非常大时) $\frac{1}{1+K_mK_c}\approx 0$

Closed-Loop Experimental Results

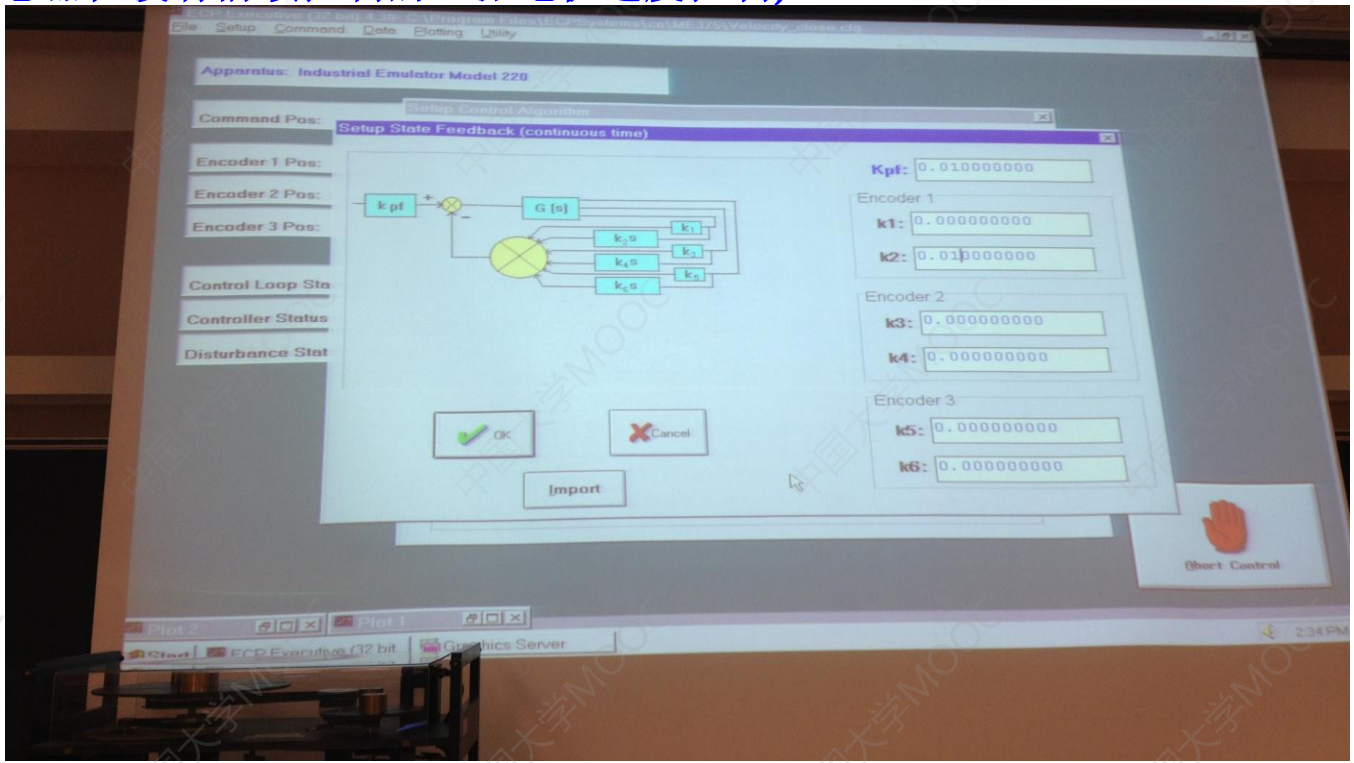
- **Speed Control of a DC Motor with Sensors but no Feedforward**

(有传感器但没有前馈控制的直流电机速度控制)



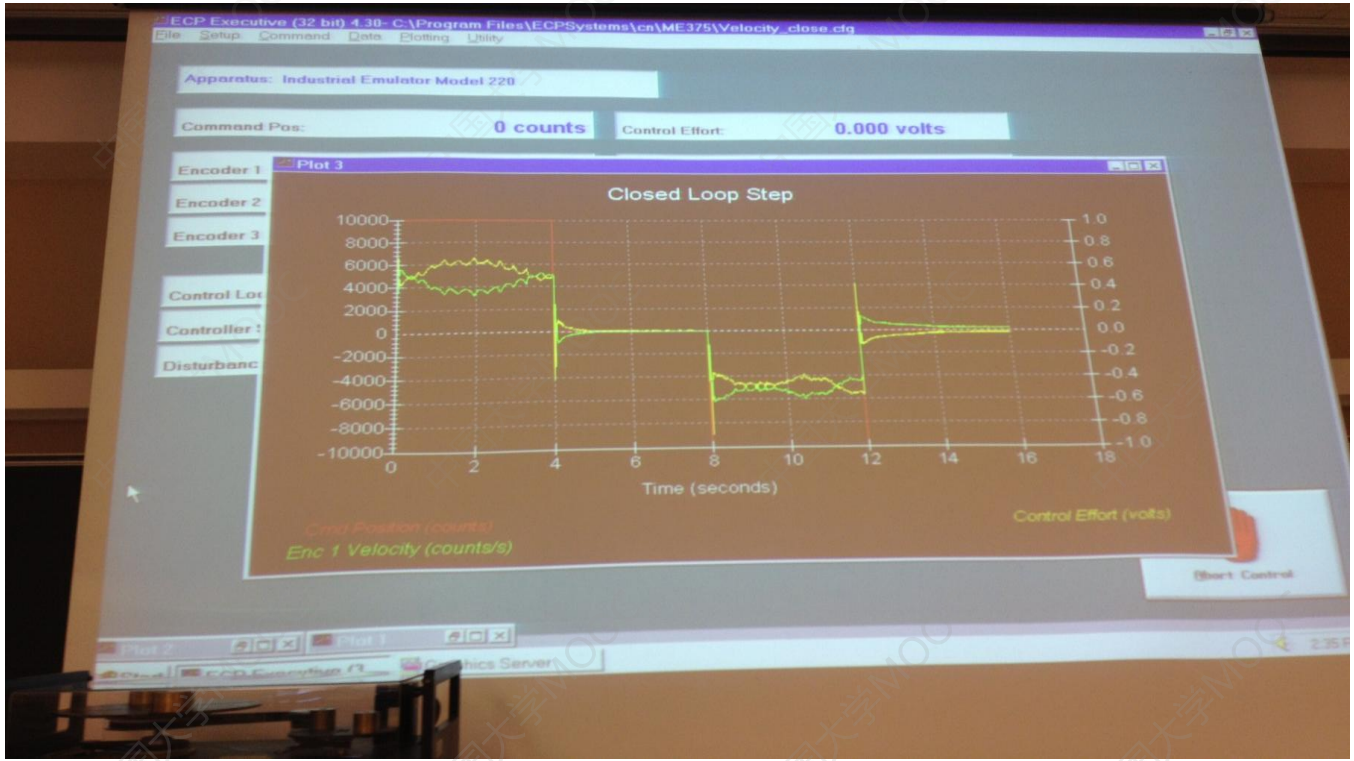
Closed-Loop Experimental Results

➢ **Speed Control of a DC Motor with Sensors but no Feedforward**
(有传感器但没有前馈控制的直流电机速度控制)



Closed-Loop Experimental Results

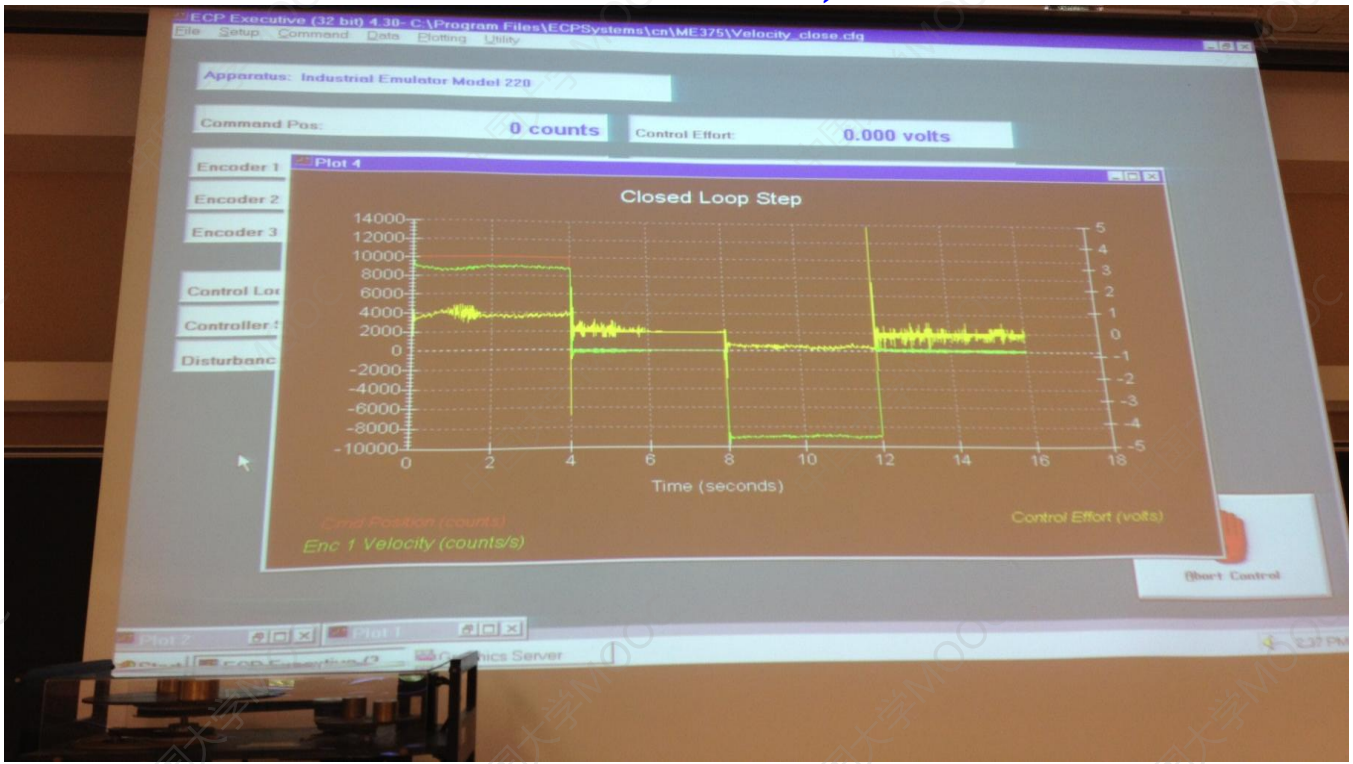
➢ **Speed Control of a DC Motor with Sensors but no Feedforward**
(有传感器但没有前馈控制的直流电机速度控制)



K_c=0.01

Closed-Loop Experimental Results

➢ **Speed Control of a DC Motor with Sensors but no Feedforward**
(有传感器但没有前馈控制的直流电机的速度控制)

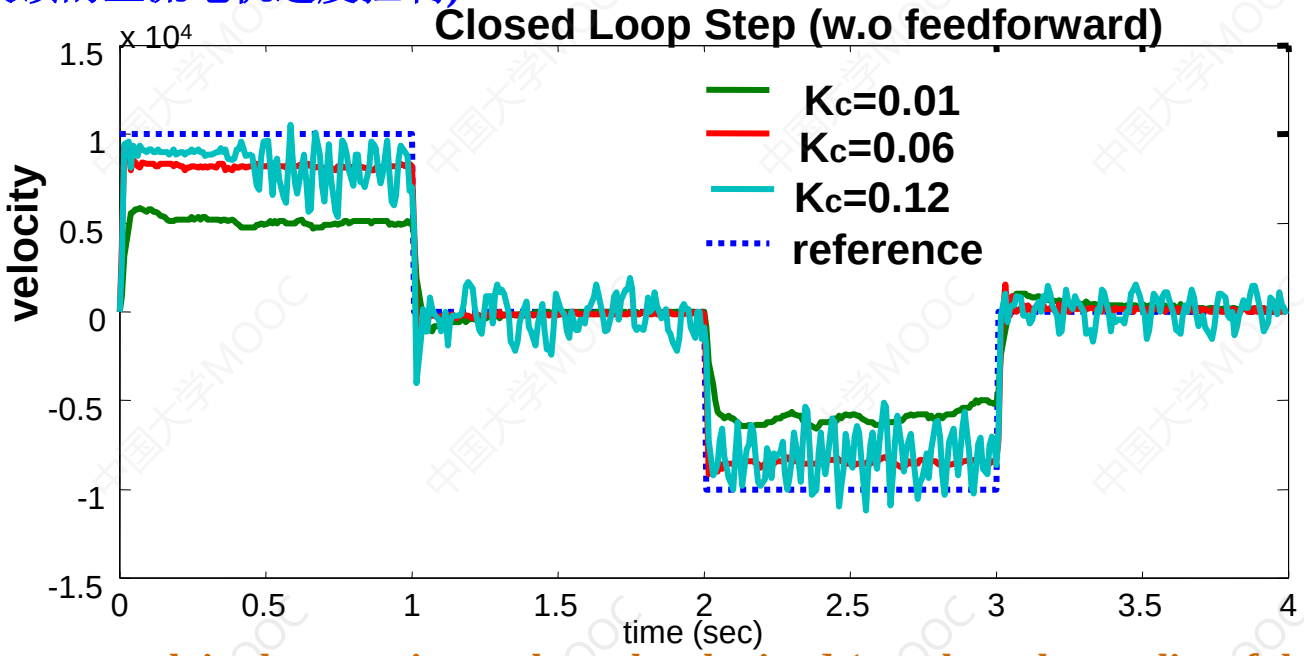


K_c=0.07

Closed-Loop Experimental Results

- **Speed Control of a DC Motor with Feedback Action Only**

(仅有反馈的直流电机速度控制)



Q: Can you explain the experimental results obtained (e.g., how the quality of the tracking performance and response speed change with feedback gain as predicted by theory)? **(你能解释所获得的实验结果吗(跟踪性能的品质和由理论预测的响应速度变化随反馈增益如何变化？)**

Summary

- **Objectives of Control System Design (控制系统设计的目标)**
- **Open-loop Control (开环控制)**
- **Close-loop Control (闭环控制)**
  - Feedback Control and Feedforward Control (反馈和前馈控制)